

Coxeter, Artin, and the Salvetti complex

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Note: The diagrams were sourced from the cited references.

1 Preliminaries

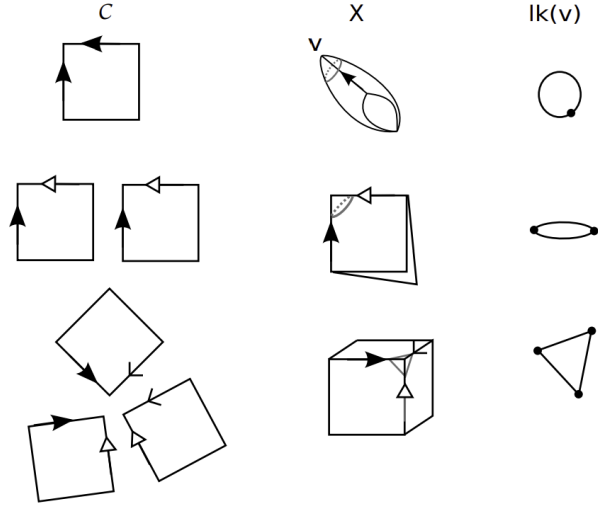
1.1 CAT(0) cube complexes

Definition 1.1 (Cube complex). Let $\mathcal{C}$ be a disjoint union of Euclidean cubes, not necessarily of the same dimension. Let $\mathcal{F}$ be a collection of isometries between faces of cubes in $\mathcal{C}$. We form a cube complex $X = \mathcal{C}/\mathcal{F}$.

Definition 1.2. A non-positively curved cube complex is a cube complex whose vertex links are simplicial flag complexes.

A 1-connected NPC complex is called a CAT(0) cube complex.

The following are not NPC's



But graphs, n -tori and closed orientable surfaces are NPC's.

Lemma 1.3. A CAT(0) cube complex is contractible.

1.2 Artin and Coxeter groups

Motivating examples

1. $B_n = \langle \delta_1, \dots, \delta_{n-1} \mid \delta_i \delta_{i+1} \delta_i = \delta_{i+1} \delta_i \delta_{i+1}, [\delta_i, \delta_j] = 1 (|i - j| \geq 2) \rangle$
2. $\Sigma_n = \langle \sigma_1, \dots, \sigma_{n-1} \mid \sigma_i^2 = 1, \sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1}, [\sigma_i, \sigma_j] = 1 (|i - j| \geq 2) \rangle$

Definition 1.4 (Artin group).

$$A = \langle S \mid R \rangle,$$

where R consists of words of the form $ststs \cdots = tstst \cdots$ ($m = m(s, t) \geq 2$, possibly ∞) for any pair of generators $s \neq t$ in S .

Examples:

1. Free groups.
2. Free abelian groups
3. B_n .

Definition 1.5 (Coxeter group).

$$W = \langle s_1, \dots, s_n \mid (s_i s_j)^{m_{ij}} = 1 \rangle,$$

where $m_{ii} = 1$ and $m_{ij} = m_{ji} \geq 2$ (possibly ∞) for $i \neq j$.

The pair (W, S) is called a Coxeter system.

Note that once we have imposed that $s_i^2 = 1$, then relations of the form $(s_i s_j)^{2m} = 1$ become equivalent to $(s_i s_j)^m (s_i s_j)^m = 1$, i.e. $(s_i s_j)^m = (s_j^{-1} s_i^{-1})^m$, i.e. $(s_i s_j)^m = (s_j s_i)^m$.

Thus a Coxeter group is a quotient of an Artin group (whose relators are words of even length) by the subgroup generated by all the squared generators.

$$1 \rightarrow \langle \langle s^2 \rangle \rangle_{s \in S} \rightarrow A \rightarrow W \rightarrow 1.$$

Hence for every Coxeter group W , there is an associated Artin group A_W .

Starting from a Coxeter system (W, S) , we form the group with generators x_s for each $s \in S$, and relations of the form

$$x_s x_t x_s \cdots = x_t x_s x_t \cdots$$

The kernel of the map $A_W \rightarrow W$, given by $x_s \mapsto s$ is called the pure Artin group PA_W .

For example, the Coxeter system $(\Sigma_n, \{\sigma_1, \dots, \sigma_{n-1}\})$ gives the Artin group B_n . The kernel of $B_n \rightarrow \Sigma_n$ is the usual PB_n .

The hidden geometry behind Coxeter groups

Let $\{H_1, \dots, H_m\}$ be a collection of hyperplanes in $\mathbb{R}^n$, and consider the group generated by reflections along these hyperplanes; say s_i along H_i . Then clearly $s_i^2 = 1$. Moreover, if H_i and H_j intersect at an angle of π/m_{ij} , then $s_i s_j$ gives a rotation by $2\pi/m_{ij}$.

Definition 1.6 (Right-angled Artin group). Let Γ be a simplicial graph, and write $V(\Gamma)$, $E(\Gamma)$ for the set of vertices and edges, respectively.

The right-angled Artin group on Γ is

$$A_\Gamma := \langle V(\Gamma) \mid [v, w] = 1 \iff \{v, w\} \in E(\Gamma) \rangle.$$

Namely, a RAAG is an Artin group for which $m_{ij} \in \{2, \infty\}$, so we can form RAAGS from Coxeter systems.

Definition 1.7 (Right-angled Coxeter group). A RACG is a quotient of a RAAG by the subgroup generated by all squared generators.

2 The Salvetti Complex

2.1 Hyperplanes

Let $\mathcal{A} = \{H_1, \dots, H_n\}$ be a collection of hyperplanes in $\mathbb{R}^n$ (called a hyperplane arrangement). These hyperplanes cut $\mathbb{R}^n$ into finitely many connected components, called chambers. We complexify the arrangement to get

$$M(\mathcal{A}) = \mathbb{C}^n - \bigcup_{H \in \mathcal{A}} H \otimes \mathbb{C}.$$

The real hyperplane arrangement gives rise to a Coxeter group W . Then W acts freely on $M(\mathcal{A})$. We get

$$\pi_1(M(\mathcal{A})/W) \cong A_W,$$

and

$$\pi_1 M(\mathcal{A}) \cong PA_W.$$

Similarly, a Coxeter system gives a hyperplane arrangement

Turns out that $M(\mathcal{A})$ has the homotopy type of a finite CW-complex, constructed purely from the combinatorics of the real hyperplane arrangement, which makes the problem of finding homotopy/homology groups manageable.

The topology and combinatorics of real hyperplane arrangements is nicely related to that of their complexification.

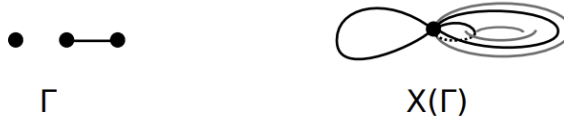
$$\#\text{chambers} = \sum_i b_i(M(\mathcal{A})),$$

and

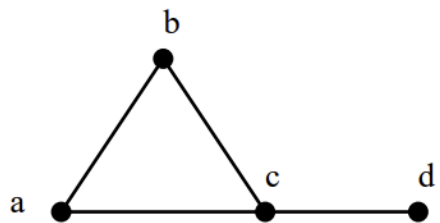
$$\#(\text{bddchambers}) = \left| \sum_i (-1)^i b_i(M(\mathcal{A})) \right|$$

Let Γ be a simplicial graph. We aim to construct a complex $\mathcal{S}_\Gamma$ such that $\pi_1(\mathcal{S}_\Gamma) \cong A_\Gamma$.

1. We start with a basepoint x , to which we attach a wedge of circles $s_1, \dots, s_n$, one for each $s_k \in V(\Gamma)$. This gives the 1-skeleton $S_\Gamma^{(1)}$.
2. For each triangle in Γ connecting the vertices s_i, s_j, s_k , attach a 3-torus with faces corresponding to the tori for the three edges of the triangle.
3. We obtain the $(n+1)$ -th skeleton inductively from $S_\Gamma^{(n)}$ as follows.
Take one copy of T^{n+1} (coming from the usual identification in an $(n+1)$ -cube) for each complete subgraph $K \subset \Gamma$ on $n+1$ vertices. We glue T^{n+1} to $S_\Gamma^{(n)}$ by identifying the torus corresponding to a subgraph of K with the subtorus of T^{n+1} spanned by the corresponding coordinates.



Example 2.1.



Example 2.2.

Theorem 2.3. Suppose that $\mathcal{A}$ is an orthogonal hyperplane arrangement of k hyperplanes. Let Γ be a simplicial graph on k vertices. Then $M(\mathcal{A})$ is homotopy equivalent to S_Γ

Proposition 2.4. The universal cover of S_Γ is a CAT(0) cube complex, so S_Γ is $K(A_\Gamma, 1)$.

More generally, we have a Salvetti complex $S_{\mathcal{A}}$ associated to a hyperplane arrangement

Why do we care? Group cohomology of a group G can be computed using a $K(G, 1)$ space.

References

- [1] Ruth Charney. *An introduction to right-angled Artin groups*. 2006. arXiv: math/0610668 [math.GR]. URL: <https://arxiv.org/abs/math/0610668>.
- [2] Michah Sageev. *CAT(0) Cube Complexes and Groups*. Lecture notes from the Park City Mathematics Institute (PCMI). Available at https://www.math.utah.edu/pcmi12/lecture_notes/sageev.pdf. 2012. URL: https://www.math.utah.edu/pcmi12/lecture_notes/sageev.pdf.